UNIVERSITY OF WESTMINSTER

INFORMATICS INSTITUTE OF TECHNOLOGY

Informatics Institute of Technology

School of Computing

Software Development II Coursework Report

Module : 4COSC010C.2: Software Development II (2023)

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Tutorial group (day, time, and tutor/s): Group 22

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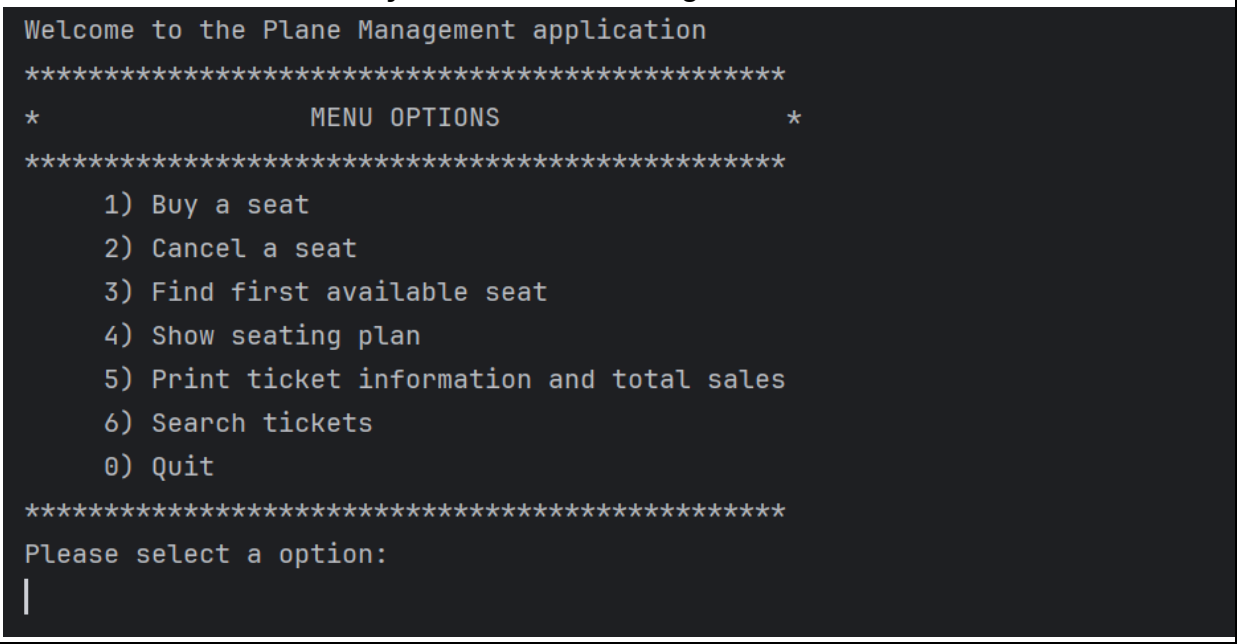
Name : Saad Mazhar

Student ID : 20221804

Self-assessment form and test plan

1) Self-assessment form

Task	Self-assessment (select one)	Comments
1	☒Fully implemented ☐Partially implemented ☐Not attempted	
2	☒Fully implemented ☐Partially implemented ☐Not attempted	
Insert here a screenshot of your welcome message and menu:		
3	☒Fully implemented ☐Partially implemented ☐Not attempted	
4	☒Fully implemented ☐Partially implemented ☐Not attempted	
5	☒Fully implemented ☐Partially implemented ☐Not attempted	
6	☐Fully implemented ☒Partially implemented	



	☐Not attempted	
Insert here a screenshot of the seating plan: Please select a option: 4 Seating Plan: A OOOOOOOOOOOOOO B OOOOOOOOOOOO C OOOOOOOOOOOO D OOOOOOOOOOOOOO Press 1 to go back to the main menu Press 0 to quit the program		
7	☒Fully implemented ☐Partially implemented ☐Not attempted	
8	☐Fully implemented ☐Partially implemented ☐Not attempted	
9	☒Fully implemented ☐Partially implemented ☐Not attempted	
10	☐Fully implemented ☒Partially implemented ☐Not attempted	
11	☐Fully implemented ☒Partially implemented ☐Not attempted	
12	☒Fully implemented ☐Partially implemented ☐Not attempted	

2) Test Plan

Complete the test plan describing which testing you have performed on your program. Add as many rows as you need.

Part A Testing

Test case / scenario	Input	Expected Output	Output	Pass/Fail
Initial state		All seats are available	Welcome message	☒Pass ☐Fail
Buy seat (valid)	1A	Seat 1 booked and updated seating plan.	Confirmation message for booking A1.	☒Pass ☐Fail
Buy Seat (invalid row)	AZ3	Error message for invalid row	Error: Invalid row, Choose A, B, C or D	☐Pass ☒Fail
Buy seat (invalid seat number)	B14	Error message for invalid seat number	Error message: Invalid seat number. Choose 1 to 12	☐Pass ☒Fail
Buy seat (already sold)	1A (after buying 1A)	Error message for seat already sold	Error message: Seat A1 is already booked.	☐Pass ☒Fail
Cancel seat (valid)	1A (after buying 1A)	Seat A1 available, updated seating plan	Confirmation message for cancelling A1. Updated seating plan with 'O' in A1	☒Pass ☐Fail
Cancel seat (invalid row)	ZA3	Error message for invalid row	Error message: Invalid row. Choose A, B, C, or D	☐Pass ☒Fail
Cancel seat (invalid seat number)	B14	Error message for invalid seat number	Error message: Invalid seat number. Choose 1 to 4	☐Pass ☒Fail
Cancel seat (already available)	1A	Error message for seat already available	Error message: Seat A1 is already available	☐Pass ☒Fail

Part B testing

Test case / scenario	Input	Expected Output	Output	Pass/Fail
Buy seat with passenger info (valid)	1A, Saad Mazhar, saad.20221804@gmail.com	Seat A1 booked with passenger info	Confirmation message for booking A1 with passenger details	☒Pass ☐Fail
Buy seat with passenger info (invalid row)	AZ3, Saad Mazhar, saad.20221804@gmail.com	Error message for invalid row	Error message: Invalid row. Choose A, B, C, or D	☐Pass ☒Fail
Cancel seat with passenger info (valid)	1A	Seat A1 available, passenger info removed	Confirmation message for cancelling A1. Ticket information removed	☒Pass ☐Fail
Cancel seat with passenger info (invalid row)	AZ3	Error message for invalid row	Error message: Invalid row. Choose A, B, C, or D	☐Pass ☒Fail
Print tickets (no tickets sold)		Message indicating no tickets sold	Message: No tickets sold in this session	☒Pass ☐Fail
Print tickets (one ticket sold)		Display ticket information for Saad Mazhar	Ticket information for John Doe (seat A1)	☒Pass ☐Fail
Search ticket (existing)	1A	Display ticket information for Saad Mazhar	Ticket information for John Doe (seat A1)	☒Pass ☐Fail
Search ticket (non-existent)	2B	Message indicating seat is available	Message: Seat B2 is available	☒Pass ☐Fail
Save ticket (valid)		Ticket information saved to file	Confirmation message for saving ticket	☒Pass ☐Fail
Save ticket (invalid)		Error message for saving failure	Error message: Error saving ticket information	☒Pass ☐Fail

Are there any specific parts of the coursework which you would like to get feedback?

You will need to demonstrate your understanding of the submitted code. Your tutor will arrange a coursework demonstration. During the coursework demonstration, your tutor will ask you to execute your program and questions on your code.

Failure to attend the demonstration will result in 0 for the coursework.

3) Code :

```
4) import java.util.InputMismatchException;
   import java.util.Scanner;
   public class W2053223_PlaneManagement {
       // Define arrays
       static int[][] seats = new int[4][14];


       static Scanner user_input = new Scanner(System.in);

       public static Ticket[] sold = new Ticket[100];
       public static int sold_count = 0;

       //main method
       public static void main(String[] args) {
           System.out.println("Welcome to the Plane Management
   application"); //welcome message
           Menu();
       }
       //menu method
       static void Menu() {

   System.out.println("************************************************");
           System.out.println("*                 MENU OPTIONS
   *");

   System.out.println("************************************************");
           System.out.println("     1) Buy a seat");
           System.out.println("     2) Cancel a seat");
           System.out.println("     3) Find first available seat");
           System.out.println("     4) Show seating plan");
           System.out.println("     5) Print ticket information and total
   sales");
           System.out.println("     6) Search tickets");
           System.out.println("     0) Quit");

   System.out.println("************************************************");
           System.out.println("Please select a option: ");
           int option = user_input.nextInt();  //get input

           if (option <= 6 && option >= 0) {   //checking if the user
   input is in the correct range
               switch (option) {
                   case 0:
                       System.out.println("exit your programme");
                       System.exit(0);
                   case 1:
                       buy_seat();  //if user input is 1
                       skip();
                       break;

                   case 2:
                       cancel_seat();    //if user input is 2
                       skip();
                       break;
```

```
                case 3:
                    find_first_available();     //if user input is 3
                    skip();
                    break;

                case 4:
                    show_seating_plan();    //if user input is 4
                    skip();
                    break;

                case 5:
                    print_tickets_info();     //if user input is 5
                    skip();
                    break;

                case 6:
                    search_tickets();     //if user input is 6
                    skip();
                    break;

            }
        }else{
            System.out.println("Enter a number between 0 and 6");
            Menu();
        }
    }


    //call buy_seat
    static void buy_seat() {
        while (true) {
            System.out.println("Enter your preferred row (A, B, C,
D)");   //ask user the preferred row
            String row_letter = user_input.next().toUpperCase();
//uppercase

            if (row_letter.matches("[ABCD]")) {    //check if input is
within the range (A,B,C or D)
                details_buy(row_letter);    //call the details_buy

                break;

            } else {
                System.out.println("Invalid row letter. Please enter A,
B, C, or D.");
            }
        }
    }


    //calling details_buy method
    static void details_buy(String row_letter) {
        int column_number;  //initialize variables
        int seatPrice;

        if (row_letter.equalsIgnoreCase("A") ||
row_letter.equalsIgnoreCase("D")) {     //check if row letter is A or B
```

```
            while (true) {
                System.out.println("Enter your preferred column (1-14):
");      //input preferred column number
                try{
                    if (user_input.hasNextInt()) {
                        column_number = user_input.nextInt();
                        if (column_number >= 1 && column_number <= 14)
{      // check the column number is in range (1 - 14)
                            break;
                        } else {
                            System.out.println("Invalid column number.
Please enter a number between 1 and 14.");
                        }
                    } else {
                        System.out.println("Invalid input. Please enter
a number.");
                        user_input.next(); // Consume the invalid input
                    }
                }catch(Exception e){
                    System.out.println("Error");
                }
            }
        } else if (row_letter.equalsIgnoreCase("B") ||
row_letter.equalsIgnoreCase("C")) {    //check if row letter is b or c
            while (true) {
                System.out.println("Enter your preferred column (1-12):
");
                if (user_input.hasNextInt()) {
                    column_number = user_input.nextInt();
                    if (column_number >= 1 && column_number <= 12) {
// check if the column number is in range (1 to 12)
                        break;
                    } else {
                        System.out.println("Invalid column number.
Please enter a number between 1 and 12.");
                    }
                } else {
                    System.out.println("Invalid input. Please enter a
number.");
                    user_input.next(); // Consume the invalid input
                }
            }
        } else {
            System.out.println("Invalid row letter");    //if it is not
equal a,b,c,d; print that
            return; // Exit the method if the row letter is invalid
        }

        user_input.nextLine();
        //input the name, surname and email from the user
        System.out.print("Enter your first name: ");
        String f_name = user_input.next();
        System.out.print("Enter your surname: ");
        String s_name = user_input.next();
        System.out.print("Enter your email: ");
        String email = user_input.next();
```

```
        //connect the person class
        Person person = new Person(f_name, s_name, email);

        // Calculate seat price based on the column number
        if (column_number <= 5) {
            seatPrice = 200;
        } else if (column_number <= 9) {
            seatPrice = 150;
        } else {
            seatPrice = 180;
        }

        int row_index = row_letter.charAt(0) - 'A';
        seats[row_index][column_number - 1] = 1;

        //ticket details
        Ticket ticket = new Ticket(row_letter, column_number,
seatPrice, person);
        sold[sold_count++] = ticket;

        ticket.printTicketInfo();

        ticket.save();     //save the ticket details
    }
    //calling cancel_seat
    static void cancel_seat() {
        Scanner user_input = new Scanner(System.in);

        System.out.println("Enter your row letter (A-D):");
        String row_letter = user_input.nextLine().toUpperCase(); //
Convert to uppercase for case-insensitive comparison

        if (!row_letter.matches("[A-D]")) { // Check for valid row
letter (A-D)
            System.out.println("Invalid row letter. Please enter A, B,
C, or D.");
            return;
        }

        System.out.println("Enter your column number:");

        int column_number;    //initialize variable
        try {
            column_number = user_input.nextInt();
        } catch (InputMismatchException e) {
            System.out.println("Invalid input. Please enter a number
for the column number.");
            user_input.nextLine(); // Consume remaining newline
character
            return;
        }

        int seat_index = row_letter.charAt(0) - 'A'; // Convert row
letter to index (A=0, B=1, ...)

        // Check if the seat is sold (assuming
seats[seat_index][column_number - 1] == 1 indicates sold seat)
```

```
        if (seats[seat_index][column_number - 1] == 1) {
            seats[seat_index][column_number - 1] = 0; // Mark seat as
available

            // Remove the canceled ticket from the sold array
            boolean ticketFound = false;
            for (int i = 0; i < sold_count; i++) {
                Ticket ticket = sold[i];
                if (ticket.get_row().equals(row_letter) &&
ticket.get_seat() == column_number) {
                    // Shift elements to remove the canceled ticket
                    for (int j = i; j < sold_count - 1; j++) {
                        sold[j] = sold[j + 1];
                    }
                    sold_count--;
                    ticketFound = true;
                    break; // Exit loop after finding a match
                }
            }
            //print cancel seat number
            if (ticketFound) {
                System.out.println("Ticket for seat " + row_letter +
column_number + " has been cancelled.");
            } else {
                System.out.println("No ticket found for seat " +
row_letter + column_number);
            }
        } else {
            System.out.println("The seat " + row_letter + column_number
+ " is available.");

        }
    }

    // calling find_first_available method
    static void find_first_available() {
        while(true){
            String[] row_letter = {"A", "B", "C", "D"};     //row
letter

            for (int i = 0; i < row_letter.length; i++) {    // It
iterates through the rows and columns of the seats array,
                String letter = row_letter[i];
                for (int j = 0; j < seats[i].length; j++) {
                    if (seats[i][j]==(0)) {
                        System.out.println("The first available seat is
" + letter + (j + 1));     // If an available seat is found, it prints
its location (row letter + seat number).
                        return;
                    }
                }
            }System.out.println("Sorry, no seats available to
select!");
        }
    }
    //calling show_seating_plan method
    static void show_seating_plan() {
```

```
        System.out.println("Seating Plan:");
        for (int i = 0; i < seats.length; i++){
            char row = (char) ('A' + i);
            System.out.print(row + " ");
            // Adjust the loop condition for printing seats based on
the row

            int seatCount = (i == 1 || i ==2) ? 12 : 14;
            for (int j = 0; j < seatCount; j++) {
                if (seats[i][j] == 1) {
                    System.out.print("X");
                } else {
                    System.out.print("O");
                }
            }
            System.out.println();

        }
    }

    //calling print ticket info method
    static void print_tickets_info() {
        while (true) {
            int total_sales = 0;
            System.out.println("Ticket information");

            // Iterate through each sold ticket
            for (int i = 0; i < sold_count; i++) {
                Ticket ticket = sold[i];

                System.out.println("Ticket " + (i + 1) + ":");
                ticket.printTicketInfo(); // Print details of the
ticket

                total_sales += ticket.get_price(); // Accumulate total
sales
            }

            // Print total sales
            System.out.println("Total sales: £" + total_sales);
            Menu();

        }
    }

    static void search_tickets() {
        while (true) {
            // Prompt the user to enter a row letter (A, B, C, D)
            System.out.println("Enter row letter (A, B, C, D): ");
            String row = user_input.next().toUpperCase();

            // Validate the row input
            if (!row.matches("[ABCD]")) {
                System.out.println("Invalid row. Please enter A, B, C,
or D.");
                return; // Exit the method
```

```
            }

            // Determine the maximum seat number based on the row
            int maxSeat = (row.equals("A") || row.equals("D")) ? 14 :
12;

            // Prompt the user to enter the seat number (1-maxSeat)
            System.out.println("Enter the seat number (1-" + maxSeat +
"): ");
            int seat_no = user_input.nextInt();

            // Validate the seat number input
            if (seat_no < 1 || seat_no > maxSeat) {
                System.out.println("Invalid seat number for row " +
row);
                return; // Exit the method
            }

            // Check if the seat is sold
            if (seats[row.charAt(0) - 'A'][seat_no - 1] == 1) {
                // Seat is sold, find the corresponding ticket
                for (Ticket ticket : sold) {
                    if (ticket.get_row().equals(row) &&
ticket.get_seat() == seat_no) {
                        System.out.println("Ticket information:");
                        ticket.printTicketInfo();
                        return; // Exit the method
                    }
                }
            } else {
                System.out.println("This seat is available.");
                Menu();
            }
        }
    }
    static void skip() {
        System.out.println("Press 1 to go back to the main menu"); //go
to the main menu
        System.out.println("Press 0 to quit the program ");  //quit the
program
        System.out.println("Enter your choice: ");
        int choice = user_input.nextInt();  //input

        if (choice == 1) {  //if 1; go to menu
            Menu();
        } else if (choice == 0) {   //if 0; end the program
            System.out.println("Goodbye!");
            System.exit(0);
        } else {
            System.out.println("Invalid choice. Exiting the program.");
//print invalid
            System.exit(0);

        }
    }
}
```

```
5) public class Person {
        // Fields
        private String f_name; // first name
        private String s_name; // surname
        private String email; // Email address

        // Constructor
        public Person(String f_name, String s_name, String email) {
            this.f_name = f_name;
            this.s_name = s_name;
            this.email = email;
        }

        // Getter for first name
        public String getFirstName() {
            return f_name;
        }

        // Setter for first name
        public void setFirstName(String f_name) {
            this.f_name = f_name;
        }

        // Getter for surname
        public String getSName() {
            return s_name;
        }

        // Setter for surname
        public void setSName(String s_name) {
            this.s_name = s_name;
        }

        // Getter for email address
        public String getEmail() {
            return email;
        }

        // Setter for email address
        public void setEmail(String email) {
            this.email = email;
        }

        // Method to print user information
        public void printPersonInfo() {
            System.out.println("Name: " + f_name);
            System.out.println("Surname: " + s_name);
            System.out.println("Email: " + email);
        }
    }
6) import java.io.File;
    import java.util.InputMismatchException;
    import java.io.*;
    public class Ticket {
        // Fields
        private String row; // The row of the seat
        private int seat; // The seat number
```

```
    private int price; // The price of the ticket
    private Person person; // Information about the ticket holder (a
Person object)

    // Constructor
    public Ticket(String row, int seat, int price, Person person) {
        this.row = row;
        this.seat = seat;
        this.price = price;
        this.person = person;
    }

    // Getter for row
    public String get_row() {
        return row;
    }

    // Setter for row
    public void set_row(String row) {
        this.row = row;
    }

    // Getter for price
    public int get_price() {
        return price;
    }

    // Setter for price
    public void set_price(int price) {
        this.price = price;
    }

    // Getter for seat
    public int get_seat() {
        return seat;
    }

    // Setter for seat
    public void set_seat(int seat) {
        this.seat = seat;
    }

    // Getter for person
    public Person get_person() {
        return person;
    }

    // Setter for person
    public void set_person(Person person) {
        this.person = person;
    }

    // Method to print ticket information
    public void printTicketInfo() {
        System.out.println("Your ticket information:");
        System.out.println("Row: " + row);
        System.out.println("Seat: " + seat);
```

```
            System.out.println("Price: £" + price);
            System.out.println("Person's information:");
            person.printPersonInfo();
        }

        // Method called save
        public void save() {
            String file_name=row + seat + ".txt";
            // print that details in text files
            try {
                //print ticket information
                FileWriter writer = new FileWriter(file_name);
                writer.write("Ticket information :\n");
                writer.write("Row letter: " + row + "\n");
                writer.write("Seat number: " + seat + "\n");
                writer.write("Price: " + price + "\n");

                //print person information
                writer.write("Person Information: \n");
                writer.write("Name is : " + person.getFirstName() + "\n");
                writer.write("Surname is : " + person.getSName() + "\n");
                writer.write("Email is : " + person.getEmail() + "\n");
                writer.close();
                System.out.println("Ticket information is in; " +
    file_name);
            }catch(IOException e){
                e.printStackTrace();

            }
        }
    }
```